清華大学 心理与认知科学系
Department of Psychological and Cognitive Sciences, Tsinghua University

GeoCGA: Geometry-Aware Cross-Modal Graph Alignment for Referring Segmentation in 3D Gaussian Splatting

Yuwen Tao, Kanglei Zhou†, Chang Li, Liyuan Wang @ Department of Psychological and Cognitive Sciences, Tsinghua University

Motivation and Key Challenges

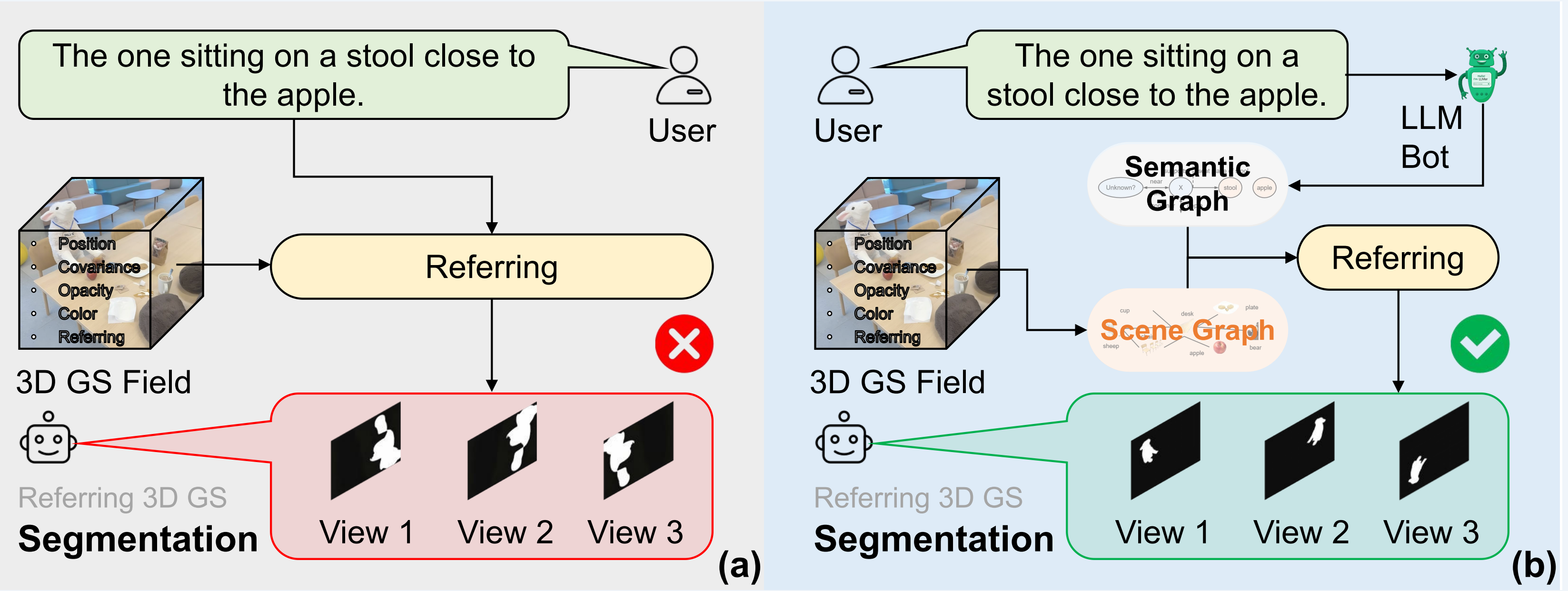
Referring 3D Segmentation seeks to localize and segment target objects in a 3D scene given a natural-language query, requiring joint reasoning over geometric structures and linguistic cues.

Two fundamental limitations remain despite advancement:

- **Lack of Positional Priors:** Language encoders provide no explicit positional encoding, weakening geometric relation modeling.
- **Self-Reinforcing Misalignment:** Cross-modal attention amplifies early alignment biases, causing spatial errors to propagate.

Limitations of existing methods:

- Rely on **global text embeddings** or direct word–feature similarity
- Fail to capture fine-grained geometric relations
- Produce **view-inconsistent** and spatially drifted predictions



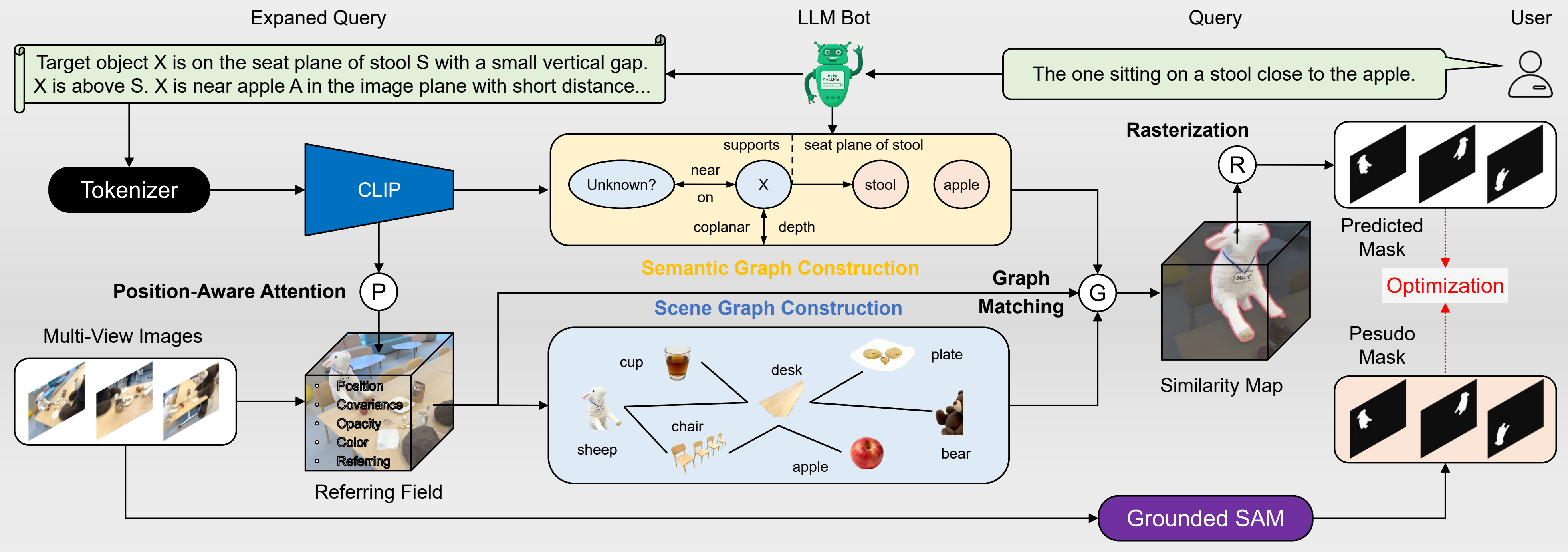
Consequences:

- Incorrect localization due to weak spatial grounding
- Coarse and spatially inconsistent segmentation masks
- Failure on relational queries

Core Idea

Instead of relying on implicit correlations, our core idea is to reformulate referring segmentation as a **geometry-aware graph alignment** problem, where both language and 3D scene representations are structured as relational graphs and aligned through **cross-modal consistency**.

Method: GeoCGA Framework



Overview of GeoCGA. GAPE constructs a semantic graph from text, while a 3D geometric graph is derived from Gaussian clusters. CMGA matches relational structures, and geometric consistency supervision enforces stable grounding.

Overview and Key Components:

- **GAPE:** Geometry-Aware Prompt Expansion enriches language with explicit spatial cues via LLM-based parsing, constructing a semantic–spatial graph
- **3DSGC:** 3D Scene Graph Construction builds an object-level geometric graph from Gaussian clusters encoding spatial topology
- **CMGA:** Cross-Modal Graph Alignment establishes fine-grained correspondence via node-level and relation-level matching
- **GCS:** Geometric Consistency Supervision enforces multi-view spatial coherence for stable grounding

Joint Training Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{align}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo}}$$

Key Advantages of GeoCGA

- Resolves **spatial ambiguity** where existing methods fail (e.g., distinguishing objects by spatial relations)
- Produces **precise masks** with clear boundaries via geometric graph reasoning
- Maintains **view-consistent** grounding across different camera perspectives

Main Results on Ref-LERF, LERF-OVS, and 3D-OVS

Table 1. Ref-LERF (mIoU ↑).

Method	**Ram.**	**Fig.**	**Tea.**	**Kit.**	**Avg.**
SPIn-NeRF	7.3	9.7	11.7	10.3	9.8
Grounded SAM	14.1	16.0	16.9	16.2	15.8
LangSplat	12.0	17.9	7.6	17.9	13.9
GS-Grouping	27.9	8.6	14.8	6.3	14.4
GOI	27.1	16.5	22.9	15.7	20.6
ReferSplat	28.3	24.3	27.2	20.1	25.0
Ours	**32.1**	**28.7**	**29.7**	**30.3**	**30.2**
Impr.	*+13.4*	*+18.1*	*+9.2*	*+50.7*	*+20.8*

Table 2. LERF-OVS (mIoU ↑).

Method	**Ram.**	**Fig.**	**Tea.**	**Kit.**	**Avg.**
Feature-3DGS	43.7	58.8	40.5	39.6	45.6
LEGaussians	46.0	60.3	40.8	39.4	46.6
LangSplat	51.2	65.1	44.7	44.5	51.4
GS-Grouping	45.5	60.9	40.0	38.7	46.3
GOI	52.6	63.7	44.5	41.4	50.6
ReferSplat	53.1	**64.1**	50.1	43.3	52.6
Ours	**56.2**	63.1	**57.0**	**46.2**	**55.6**
Impr.	*+5.8*	*–1.6*	*+13.8*	*+6.7*	*+5.7*

Table 3. Comparison on **3D-OVS** (mIoU ↑). Top methods shown.

Method	**Venue**	**Bed**	**Bench**	**Room**	**Sofa**	**Lawn**	**Average**
LangSplat	CVPR'24	92.5	94.2	94.1	90.0	96.1	93.4
GOI	MM'24	89.4	92.8	91.3	85.6	94.1	90.6
ReferSplat	ICML'25	**90.2**	93.8	94.1	90.8	95.5	92.9
GeoCGA (Ours)	–	90.1	**95.4**	**93.7**	**93.4**	**96.0**	**93.7**
Improv. (%)	–	*–0.1*	*+1.7*	*–0.4*	*+2.9*	*+0.5*	*+1.0*

- GeoCGA achieves **+20.8%** mIoU on Ref-LERF, **+5.7%** on LERF-OVS, and **+1.0%** on 3D-OVS

Ablation and Computation

Table 4. Ablation study.

Setting	**Sem.**	**Geo.**	**Ramen**	**Kitchen**
Baseline 0	×	×	28.3	20.1
Baseline 1	✓	×	$29.5^{+1.2}$	$23.8^{+2.7}$
Baseline 2	×	✓	$30.4^{+2.1}$	$26.5^{+6.4}$
Ours	✓	✓	$\mathbf{32.1}^{+3.8}$	$\mathbf{30.3}^{+10.2}$

Table 5. Computation.

	ReferSplat	**Ours**
Params (M)	304.18	**128.48**
FLOPs (G)	41.82	**25.28**
Ref-LERF	0.0	**+20.8**
LERF-OVS	0.0	**+5.7**
3D-OVS	0.0	**+1.0**

Conclusion, Limitation, and Future Work

- GeoCGA achieves SOTA on three benchmarks by bridging linguistic semantics with 3DGS geometry via structured graph alignment, and the future can be towards open-vocabulary 4D reasoning